

SANTA MONICA, CA, USA

WMO: 722885

Lat: 34.016N

Lon: 118.451W

Elev: 53

StdP: 100.69

Time Zone: -8.00 (NAP)

Period: 00-19

WBAN: 93197

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	6.3	7.4	-12.9	1.2	17.3	-9.7	1.7	16.2	7.5	14.3	6.7	14.0	1.5	40	0.349

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.1	29.8	16.7	27.8	17.0	26.3	17.2	21.3	25.2	20.6	24.5	19.9	23.8	2.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.8	14.6	23.4	19.0	13.9	22.9	18.2	13.2	22.1	62.0	25.3	59.3	24.6	57.1	23.8	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.5	5.6	5.0	4.2	34.5	1.1	2.6	3.4	36.3	2.8	37.9	2.1	39.3	1.3	41.2
			0.3	21.9	1.3	1.2	-0.6	22.8	-1.3	23.5	-2.0	24.1	-3.0	25.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.2	14.3	13.9	15.0	15.8	17.1	18.7	20.6	20.8	20.6	18.9	16.4	13.9	(6)
(7)		DBStd	3.57	3.04	2.81	2.54	2.50	2.18	1.51	1.88	1.90	2.56	2.83	2.96	2.51	(7)
(8)		HDD10.0	4	2	1	0	0	0	0	0	0	0	0	0	1	(8)
(9)		HDD18.3	761	129	128	109	86	51	11	2	2	4	25	72	141	(9)
(10)		CDD10.0	2628	136	110	156	173	219	261	328	335	319	277	194	121	(10)
(11)		CDD18.3	344	6	4	7	9	12	22	72	78	74	44	16	2	(11)
(12)		CDH23.3	1375	53	39	57	71	80	39	147	200	293	261	119	16	(12)
(13)	CDH26.7	320	7	8	13	20	31	8	16	22	73	89	32	1	(13)	
(14)	Wind	WSAvg	2.1	1.9	2.1	2.2	2.4	2.3	2.3	2.1	2.0	1.9	1.8	1.9	(14)	
(15)	Precipitation	PrecAvg	403	91	107	55	23	10	2	1	5	17	28	61	(15)	
(16)		PrecMax	750	406	371	178	61	102	21	8	2	59	138	67	246	(16)
(17)		PrecMin	166	6	3	0	1	0	0	0	0	0	0	0	1	(17)
(18)		PrecStd	186	100	95	54	17	20	5	2	1	12	28	21	59	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.6	27.9	28.4	29.4	30.9	27.7	28.5	28.8	32.3	32.9	30.2	25.6	(19)
(20)			MCWB	13.2	13.2	14.3	14.0	14.4	19.2	20.8	20.1	19.6	16.7	14.4	11.9	(20)
(21)		2%	DB	24.7	23.8	24.5	25.0	24.8	23.9	26.3	27.0	28.5	29.0	26.9	22.6	(21)
(22)			MCWB	11.7	12.0	12.9	13.3	15.1	18.1	20.0	19.8	19.6	15.5	13.6	11.4	(22)
(23)		5%	DB	22.4	21.1	21.6	22.1	22.0	22.4	24.8	25.5	26.4	26.2	23.9	20.3	(23)
(24)			MCWB	11.4	11.8	12.6	13.0	15.8	17.5	19.6	19.6	19.2	15.8	12.7	11.2	(24)
(25)	10%	DB	19.8	18.4	19.1	19.8	20.2	21.4	23.7	24.1	24.6	23.5	21.4	18.3	(25)	
(26)		MCWB	10.7	11.6	12.7	13.3	15.2	17.0	19.2	19.2	18.9	16.6	12.9	11.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.6	15.7	16.6	16.6	18.0	22.0	22.0	22.2	20.9	17.3	16.1	(27)	
(28)			MCDWB	19.3	19.7	22.5	22.8	23.3	25.3	25.6	25.9	27.2	25.3	20.8	17.9	(28)
(29)		2%	WB	14.7	14.7	15.5	15.5	17.0	18.7	21.1	21.0	21.0	19.4	16.4	15.0	(29)
(30)			MCDWB	18.0	18.0	20.1	19.9	21.5	23.0	24.4	25.1	25.5	24.0	20.2	17.8	(30)
(31)		5%	WB	14.0	14.0	14.7	14.8	16.3	18.0	20.4	20.2	20.3	18.5	15.7	14.3	(31)
(32)			MCDWB	17.4	17.1	18.5	18.9	20.3	21.7	23.7	24.2	24.6	22.6	19.5	17.1	(32)
(33)	10%	WB	13.3	13.3	14.1	14.2	15.7	17.5	19.7	19.5	19.5	17.7	15.1	13.5	(33)	
(34)		MCDWB	16.7	16.7	17.3	18.1	19.3	20.7	23.1	23.4	23.6	21.1	19.0	16.3	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.2	8.5	7.7	7.3	6.1	5.5	5.7	6.1	6.9	7.9	9.1	8.8	(35)
(36)			MCDBR	13.4	13.1	12.1	11.7	9.2	7.2	7.1	7.6	9.6	12.9	13.3	12.1	(36)
(37)			MCWBR	6.4	6.4	5.7	5.3	4.2	3.3	3.1	3.4	4.1	6.1	6.7	6.6	(37)
(38)		5% WB	MCDWB	8.8	8.7	8.0	8.1	7.0	6.8	6.6	6.8	8.0	8.7	9.1	8.2	(38)
(39)			MCWBR	5.1	5.0	4.4	4.1	3.4	3.2	2.9	3.2	3.6	4.8	5.4	5.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.332	0.345	0.368	0.378	0.385	0.400	0.404	0.383	0.371	0.372	0.321	0.323	(40)	
(41)		taud	2.504	2.480	2.395	2.367	2.378	2.342	2.393	2.433	2.461	2.430	2.570	2.562	(41)	
(42)		Ebn at Noon	878	904	906	909	901	883	877	893	891	860	887	870	(42)	
(43)		Edh at Noon	85	96	113	122	122	126	119	113	105	100	79	76	(43)	
(44)	All-Sky Solar Radiation	RadAvg	3.14	4.02	5.39	6.61	7.16	7.48	7.47	7.10	5.99	4.54	3.50	2.85	(44)	
(45)		RadStd	0.23	0.41	0.31	0.35	0.49	0.46	0.40	0.24	0.24	0.34	0.21	0.22	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.83	N/A	-1.61	+1.14	+0.95	N/A	N/A	-154	+294	+161
(47)	Regional (52 neighbors)	+0.64	N/A	-1.65	+0.73	N/A	N/A	N/A	-118	+225	+108

Nomenclature: See separate page